



United International University

School of Science and Engineering

Department of Computer Science and Engineering

Final Examination || Trimester: Fall – 2025

Course Title: Probability & Statistics || Code: Math -2205

Total marks – 40 || Duration – 2 hours

(Note that the number of marks is given in brackets [] at the end of each question or part question. You are requested to answer all the questions in order.)

- Q1** (a) A public health department wants to estimate the average daily screen time (in hours) of university students. A random sample of 64 students is selected, and the sample shows a mean screen time of 5.2 hours with a standard deviation of 1.6 hours. Construct a 90% confidence interval for the population mean of daily screen time of university students. [3]
- (b) A factory produces rechargeable batteries that are advertised to have an average lifespan of more than 80 hours. To verify this claim, a quality control analyst randomly selects 16 batteries from the production line. The test results show that the batteries have an average lifespan of 82 hours with a standard deviation of 4 hours. Assuming that battery lifespans follow a normal distribution, test at the 5% level of significance whether there is sufficient evidence to support the manufacturer's claim that the population mean lifespan exceeds 80 hours. [5]
- (c) For a two-tailed Z-test at 5% level of significance, sketch the critical region. [2]
- Q2** a) Given that $P(A) = 0.6$, $P(B) = 0.4$. If A and B are independent, find $p(A \cup B)$. [2]
- b) A hospital uses a medical test to detect a certain disease. 2% of the population has the disease. The test correctly identifies the disease 90% of the time when a person truly has it. The test incorrectly gives a positive result 5% of the time when a person does *not* have the disease. If a randomly selected person tests positive, what is the probability that the person actually has the disease? [3]
- c) An e-commerce company tracks the number of days a delivery is delayed beyond the promised date. The probability distribution of the delay X (in days) is given below:
- | | | | | |
|----------------------------------|----------|----------|----------|----------|
| Delay days x | 0 | 1 | 2 | 3 |
| $P(X = x)$ | 0.40 | 0.30 | 0.20 | 0.10 |
- (i) Verify that the given distribution is a valid probability distribution. [1]
- (ii) Find the mean (expected delay). [2]
- (iii) Find the variance of the delay [2]
- Q3** (a) The daily electricity consumption of households in a city follows a normal distribution with a mean of 30 kWh and a standard deviation of 5 kWh. Determine the proportion of households whose daily consumption lies between 25 kWh and 35 kWh [4]
- (b) A mobile network provider knows that 2% of its customers experience a drop during peak hours. If 20 customers are randomly selected during peak time, find the probability that at least one customer experiences a call drop. [3]

P.T.O.

- (c) On average, 3 customer complaints are received per hour by a customer service center. Find the probability that **no more than 1 complaints** will be received in a **30-minute period**. [3]

Q4 At a bank, the time (in minutes) required to serve a customer at a counter is modeled by the following probability density function (PDF):

$$f(t) = \begin{cases} k t(10 - t), & 0 \leq t \leq 10 \\ 0, & \text{otherwise} \end{cases}$$

- (i) Find the value of k so that $f(t)$ is a valid probability density function. [2]
(ii) Obtain the cumulative distribution function (CDF) $F(t)$. [3]
(iii) Find the probability that a customer is served in less than 4 minutes. [3]
(iv) Determine the mode. [2]

Related Formulae

	<u>pmf/pdf</u>
Binomial	$f(x) = n_c p^x (1 - p)^{n-x}; x = 0, 1, 2, \dots, n$
Poisson	$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}; x = 0, 1, 2, \dots$
Confidence Interval	<i>For Population Mean</i> $\bar{x} - z \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z \frac{\sigma}{\sqrt{n}}$ <i>For population proportion</i> $\hat{p} - z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} < p < \hat{p} + z \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Normal	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}; -\infty < x < \infty$